

THEORY AND METHODS

GENERALIZATION OF CRAMER'S COEFFICIENT OF ASSOCIATION FOR CONTINGENCY TABLES

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Summary: For the analysis of two-way contingency tables, Cramer (1946, p. 282) considered the measure V^2 which describes the strength of association between the row and column variables, namely the degree of departure from independence. This paper proposes three kinds of measures which describe strength of association, by applying the Cressie and Read's (1984) power-divergence. The proposed measures include Cramer's V^2 , and Theil's (1970) uncertainty coefficient as special cases. The measures would be useful for comparing the strength of association in several tables. Examples are given.

1. Introduction

For an $r \times c$ contingency table, let p_{ij} denote the probability that an observation will fall in the i th row and j th column of the table ($i = 1, 2, \dots, r$; $j = 1, 2, \dots, c$). Many coefficients have been proposed to measure the strength of association between the row and column variables, namely to measure the degree of departure from independence. Examples include Pearson's coefficients ϕ^2 of mean square contingency and P of contingency, Tshuprow's coefficient T , and Cramer's (1946, p. 282) coefficient V^2 (see, e.g., Bishop, STMA (2004) SUBJECT CLASSIFICATION: 06:030(05:110)

Fienberg and Holland, 1975, Ch.11; Everitt, 1992, p. 54; Agresti, 1990, p. 75). The measure ϕ^2 is defined by

$$\begin{aligned}\phi^2 &= \sum_{i=1}^r \sum_{j=1}^c \frac{(p_{ij} - p_{i.} p_{.j})^2}{p_{i.} p_{.j}} \\ &= \left(\sum_{i=1}^r \sum_{j=1}^c \frac{p_{ij}^2}{p_{i.} p_{.j}} \right) - 1,\end{aligned}$$

where $p_{i.} = \sum_{j=1}^c p_{ij}$ and $p_{.j} = \sum_{i=1}^r p_{ij}$. When $r > 2$ or $c > 2$, this measure does not lie between 0 and 1. The measure P , defined by

$$P = \left(\frac{\phi^2}{\phi^2 + 1} \right)^{1/2},$$

lies between 0 and 1 but it cannot always attain the upper limit of 1. We note that even under complete association, the maximum value of P depends on the number of rows and columns in the table. To avoid these limitations of P , the measure T is defined by

$$T = \left(\frac{\phi^2}{[(r-1)(c-1)]^{1/2}} \right)^{1/2}.$$

This attains the value 1 in the case of complete association in an $r \times r$ table, but cannot do so when $r \neq c$. Cramer's V^2 , which attains the value 1 for all values of r and c in the case of complete association, is defined by

$$V^2 = \frac{\phi^2}{\min(r-1, c-1)}.$$

We note that $0 \leq V^2 \leq 1$, and when $r = c$, then $V = T$, but otherwise $V > T$.

Let n_{ij} denote the observed frequency in the i th row and j th column of the table ($i = 1, 2, \dots, r; j = 1, 2, \dots, c$) and let $n = \sum_{i=1}^r \sum_{j=1}^c n_{ij}$. Pearson's chi-squared statistic X^2 for testing independence between the row

and column variables is given by

$$X^2 = \sum_{i=1}^r \sum_{j=1}^c \frac{(n_{ij} - n_{i.}n_{.j}/n)^2}{n_{i.}n_{.j}/n},$$

where $n_{i.} = \sum_{j=1}^c n_{ij}$ and $n_{.j} = \sum_{i=1}^r n_{ij}$. We note that $X^2/n = \hat{\phi}^2$, where $\hat{\phi}^2$ is given by ϕ^2 with $\{p_{ij}\}$ replaced by $\{\hat{p}_{ij}\}$, where $\hat{p}_{ij} = n_{ij}/n$. Also, Cressie and Read's (1984) power-divergence statistic for testing independence is given by

$$W_{(\lambda)}^2 = \frac{2}{\lambda(\lambda+1)} \sum_{i=1}^r \sum_{j=1}^c n_{ij} \left[\left(\frac{n_{ij}}{n_{i.}n_{.j}/n} \right)^\lambda - 1 \right] \quad \text{for } -\infty < \lambda < \infty,$$

where the cases $\lambda = 0$ and $\lambda = -1$ are defined as the limits as $\lambda \rightarrow 0$ and $\lambda \rightarrow -1$, respectively. We note that $W_{(0)}^2$ is identical to the likelihood ratio chi-squared statistic (denoted by G^2) and that $W_{(-1)}^2$ is identical to X^2 (see also Read and Cressie, 1988, p. 15). So, we are interested in a measure analogous to Cramer's V^2 by applying the power-divergence.

The purpose of this paper is to propose power-divergence type measures which represent strength of association. The proposed measures are generalizations of Cramer's V^2 and reduce to Theil's (1970) uncertainty coefficient in a special case. The measures would be useful for comparing the strength of association between the row and column variables in several tables.

2. Measures of association

Consider an $r \times c$ contingency table. We assume that $\{p_{i.} \neq 0, p_{.j} \geq 0\}$ when $r \leq c$, and $\{p_{i.} \geq 0, p_{.j} \neq 0\}$ when $r > c$. We shall propose measures which represent the strength of association between the row and column variables in three cases: (I) when the row and column variables are response and explanatory variables respectively, (II) when those

are explanatory and response variables respectively and (III) when explanatory and response variables are not defined. We shall define measures for the asymmetric situation (in the case of (I) and (II)) and for the symmetric situation (in the case of (III)).

2.1 Case I

Consider the asymmetric situation in which the column variable is the explanatory variable and the row variable is the response variable. Then we consider the following measure which represents the strength of association between row and column variables:

$$V_{1(\lambda)}^2 = \frac{I^{(\lambda)}(\{p_{ij}\}; \{p_{i \cdot} p_{\cdot j}\})}{K_1^{(\lambda)}} \quad \text{for } \lambda \geq 0,$$

where

$$I^{(\lambda)}(\cdot; \cdot) = \frac{1}{\lambda(\lambda+1)} \sum_{i=1}^r \sum_{j=1}^c p_{ij} \left[\left(\frac{p_{ij}}{p_{i \cdot} p_{\cdot j}} \right)^\lambda - 1 \right],$$

$$K_1^{(\lambda)} = \frac{1}{\lambda(\lambda+1)} \left(\sum_{i=1}^r p_{i \cdot}^{1-\lambda} - 1 \right),$$

and the value at $\lambda = 0$ is taken to be the continuous limit as $\lambda \rightarrow 0$. Note that λ is a real value that is chosen by the user. When $\lambda = 0$, this measure is expressed as

$$V_{1(0)}^2 = \frac{\sum_{i=1}^r \sum_{j=1}^c p_{ij} \log \left(\frac{p_{ij}}{p_{i \cdot} p_{\cdot j}} \right)}{- \sum_{i=1}^r p_{i \cdot} \log p_{i \cdot}}.$$

Note that $I^{(\lambda)}(\cdot; \cdot)$ is the power-divergence between two distributions $\{p_{ij}\}$ and $\{p_{i \cdot} p_{\cdot j}\}$ which includes as special cases the Kullback-Leibler information (when $\lambda = 0$) and one half the Pearson chi-squared type discrepancy (when $\lambda = 1$). For more details on power-divergence, see Cressie and Read (1984), and Read and Cressie (1988, p. 15). We also note that $K_1^{(0)}$ is the Shannon entropy for the row marginal distribution $\{p_{i \cdot}\}$. We point out that

when $\lambda = 1$ with $r \leq c$, the measure $V_{1(1)}^2$ is identical to Cramer's V^2 . In addition, we note that when $\lambda = 0$, the measure $V_{1(0)}^2$ is identical to Theil's (1970) uncertainty coefficient (see Agresti, 1990, p. 25), which describes the proportional reduction in variation from the row marginal distribution to the conditional distributions of the row response (though the detail is omitted here).

We note that the power-divergence $I^{(\lambda)}(\{p_{ij}\}; \{p_{i \cdot} p_{\cdot j}\})$ is nonnegative and is zero if and only if $\{p_{ij} = p_{i \cdot} p_{\cdot j}\}$ hold, namely, there is a structure of null association in the table (i.e. independence between the row and column variables). Moreover, we note that the inequalities $p_{ij} \leq p_{\cdot j}$ hold with $p_{ij} \geq 0$ when $p_{\cdot j} > 0$. Thus, noting that $f_1(x) = x^\lambda$ for $\lambda > 0$ and $f_2(x) = \log x$ are monotonic increasing functions of $x(> 0)$ and defining $0(\frac{0}{0})^\lambda = 0(\lambda > 0)$ and $0 \log 0 = 0$, we see that when $\lambda > 0$,

$$\begin{aligned} I^{(\lambda)}(\cdot; \cdot) &\leq \frac{1}{\lambda(\lambda+1)} \left[\sum_{i=1}^r \sum_{j=1}^c p_{ij} \left[\left(\frac{1}{p_{i \cdot}} \right)^\lambda - 1 \right] \right] \\ &= \frac{1}{\lambda(\lambda+1)} \left(\sum_{i=1}^r p_{i \cdot}^{1-\lambda} - 1 \right), \end{aligned}$$

and when $\lambda = 0$,

$$\begin{aligned} I^{(0)}(\cdot; \cdot) &\leq \sum_{i=1}^r \sum_{j=1}^c p_{ij} \log \left(\frac{1}{p_{i \cdot}} \right) \\ &= - \sum_{i=1}^r p_{i \cdot} \log p_{i \cdot}. \end{aligned}$$

Therefore, we obtain that $0 \leq V_{1(\lambda)}^2 \leq 1$ for $\lambda \geq 0$, and for each λ , (I) $V_{1(\lambda)}^2 = 0$ if and only if there is a structure of null association in the table, i.e. $\{p_{ij} = p_{i \cdot} p_{\cdot j}\}$, and (II) $V_{1(\lambda)}^2 = 1$ if and only if there is a structure of complete association in the table; namely (i) for each column j ($j = 1, 2, \dots, c$), $p_{i_j, j} > 0$ for one i_j (note that $i_m = i_l$, $m < l$, may hold when $r \neq c$) and $p_{ij} = 0$ for the other all $i(\neq i_j)$ (or $p_{ij} = 0$ for all i). We point out

that when $V_{1(\lambda)}^2 = 1$ with $r > c$ and $\lambda \neq 1$, the row marginal probabilities $\{p_{i.}\}$ in $K_1^{(\lambda)}$ are $p_{ik.} > 0$ for $k = 1, 2, \dots, c$ but $p_{ik.} = 0$ for the other $k = c + 1, c + 2, \dots, r$, and so only when $\lambda = 1$ with $r > c$, the $V_{1(1)}^2$ does not attain to 1.

For each λ , when $V_{1(\lambda)}^2 = 1$ we can predict the row category of an individual perfectly when we know the column category and it is impossible to predict it perfectly when we do not. In contrast, when $V_{1(\lambda)}^2 = 0$, then the row marginal distribution $\{p_{i.}\}$ is identical to the conditional row marginal distribution $\{p_{ij}/p_{.j}\}$ on condition that the value of column category is given. Therefore, when $V_{1(\lambda)}^2 = 0$, then the column category of the individual is of no use in predicting the row category. So, the measure $V_{1(\lambda)}^2$ would indicate to what degree the prediction of the row category of an individual may be improved if knowledge regarding the column category is available. In this sense, the measure $V_{1(\lambda)}^2$ indicates the strength of association between row and column variables. Also, the measure $V_{1(\lambda)}^2$ describes the relative magnitude of the degree of association between row and column variables to the degree of complete association (equals $K_1^{(\lambda)}$), which depends only on the row marginal distribution $\{p_{i.}\}$.

2.2 Case II

Consider the asymmetric situation in which the row variable is the explanatory variable and the column variable is the response variable. Then we consider the following measure which represents the strength of association between row and column variables:

$$V_{2(\lambda)}^2 = \frac{I^{(\lambda)}(\{p_{ij}\}; \{p_{i.}p_{.j}\})}{K_2^{(\lambda)}} \quad \text{for } \lambda \geq 0,$$

where

$$K_2^{(\lambda)} = \frac{1}{\lambda(\lambda+1)} \left(\sum_{j=1}^c p_{.j}^{1-\lambda} - 1 \right),$$

and the value at $\lambda = 0$ is taken to be the continuous limit as $\lambda \rightarrow 0$. When $\lambda = 0$, the measure is expressed as

$$V_{2(0)}^2 = \frac{\sum_{i=1}^r \sum_{j=1}^c p_{ij} \log \left(\frac{p_{ij}}{p_{i.} p_{.j}} \right)}{- \sum_{j=1}^c p_{.j} \log p_{.j}}.$$

Note that when $\lambda = 1$, the measure $V_{2(1)}^2$ with $r \geq c$ is identical to Cramer's V^2 . When $\lambda = 0$, the measure $V_{2(0)}^2$ is identical to Theil's (1970) uncertainty coefficient which describes the proportional reduction in variation from the column marginal distribution to the conditional distributions of the column response.

We note that $0 \leq V_{2(\lambda)}^2 \leq 1$ for $\lambda \geq 0$, and for each λ , $V_{2(\lambda)}^2 = 0$ if and only if there is a structure of null association in the table, i.e. $\{p_{ij} = p_{i.} p_{.j}\}$. Also, $V_{2(\lambda)}^2 = 1$ if and only if there is a structure of complete association in the table; namely for each row i ($i = 1, 2, \dots, r$), $p_{i,j_i} > 0$ for one j_i (note that $j_m = j_l$, $m < l$, may hold when $r \neq c$) and $p_{ij} = 0$ for all other j ($\neq j_i$) [or $p_{ij} = 0$ for all j]. We point out that when $V_{2(\lambda)}^2 = 1$ with $r < c$ and $\lambda \neq 1$, the column marginal probabilities $\{p_{.j}\}$ in $K_2^{(\lambda)}$ are $p_{.j_k} > 0$ for $k = 1, 2, \dots, r$, but $p_{.j_k} = 0$ for the other $k = r+1, r+2, \dots, c$. Thus, only when $\lambda = 1$ with $r < c$, $V_{2(1)}^2$ cannot attain the value 1.

Similar to the interpretation of $V_{1(\lambda)}^2$, the measure $V_{2(\lambda)}^2$ indicates to what degree prediction of the column category of an individual may be improved if knowledge regarding the row category of the individual is available. In this sense, the measure $V_{2(\lambda)}^2$ indicates the strength of association between row and column variables. Also, the measure $V_{2(\lambda)}^2$ describes the relative magnitude of

the degree of association between the row and column variables to the degree of complete association (equals $K_2^{(\lambda)}$), which depends only on the column marginal distribution $\{p_{\cdot j}\}$.

2.3 Case III

Consider an $r \times c$ contingency table in which explanatory and response variables are not defined. For such tables, it would not be suitable to use the measures $V_{1(\lambda)}^2$ and $V_{2(\lambda)}^2$ if one wants to see what degree knowledge of the value of one variable is useful for predicting the value of the other variable. For such a symmetric situation, we shall consider the following measure by combining the ideas of both measures $V_{1(\lambda)}^2$ and $V_{2(\lambda)}^2$:

$$V_{3(\lambda)}^2 = \frac{2I^{(\lambda)}(\{p_{ij}\}; \{p_{i\cdot} p_{\cdot j}\})}{K_1^{(\lambda)} + K_2^{(\lambda)}} \quad \text{for } \lambda \geq 0,$$

where the value at $\lambda = 0$ is taken to be the continuous limit as $\lambda \rightarrow 0$. Namely, when $\lambda = 0$, the measure is expressed as

$$V_{3(0)}^2 = \frac{2 \sum_{i=1}^r \sum_{j=1}^c p_{ij} \log \left(\frac{p_{ij}}{p_{i\cdot} p_{\cdot j}} \right)}{- \sum_{i=1}^r p_{i\cdot} \log p_{i\cdot} - \sum_{j=1}^c p_{\cdot j} \log p_{\cdot j}}. \quad (2.1)$$

We point out that when $\lambda = 1$ with $r = c$, the measure $V_{3(1)}^2$ equals $V_{1(1)}^2$ (and $V_{2(1)}^2$) which is identical to Cramer's V^2 . Also, when $\lambda = 0$, the measure $V_{3(0)}^2$ in (2.1) is identical to the total uncertainty measure (see Freeman, 1987, p. 101).

We note that for each λ , $0 \leq V_{3(\lambda)}^2 \leq 1$ and that $V_{3(\lambda)}^2 = 0$ if and only if there is independence while $V_{3(\lambda)}^2 = 1$ if and only if there is a structure of complete association in the table; namely (i) when $r \leq c$, for each row i ($i = 1, 2, \dots, r$), $p_{i,j_i} > 0$ for one j_i , where $j_m \neq j_l$, $m < l$, and $p_{ij} = 0$ for the other all $j (\neq j_i)$, and (ii) when $r > c$, for each column j ($j = 1, 2, \dots, c$),

$p_{i_j, j} > 0$ for one i_j , where $i_m \neq i_l$, $m < l$, and $p_{ij} = 0$ for all other $i (\neq i_j)$. Note that only when $\lambda = 1$ with $r \neq c$, $V_{3(1)}^2$ cannot attain the value 1. Especially, for an $r \times r$ square contingency table with the same row and column classifications, $V_{3(\lambda)}^2 = 1$ if and only if the main diagonal cell probabilities in the $r \times r$ table are nonzero and the off-diagonal cell probabilities are all zero (so, all observations concentrate on the main diagonal cells) after interchanging some row and column categories.

Therefore, in predicting the values of categories of an individual, the measure $V_{3(\lambda)}^2$ indicates to what degree prediction may be improved if knowledge regarding the value of one variable is available. In such a sense the measure $V_{3(\lambda)}^2$ also indicates the strength of association between the row and column variables. Consider predicting the values of row and column categories of an individual in terms of probabilities with independence structure when we know only the marginal distributions $\{p_i\}$ and $\{p_j\}$. Then the measure $V_{3(\lambda)}^2$ indicates to what degree prediction is improved when we know the full distribution $\{p_{ij}\}$. Also, the measure $V_{3(\lambda)}^2$ describes the relative magnitude of the degree of association between the row and column variables to the degree of complete association (equals $(K_1^{(\lambda)} + K_2^{(\lambda)})/2$), which depends only on the row and column marginal distributions $\{p_i\}$ and $\{p_j\}$.

We note that the interpretation of the measure $V_{3(\lambda)}^2$ is different from those of the measures $V_{1(\lambda)}^2$ and $V_{2(\lambda)}^2$, except in the case $\lambda = 1$ with $r = c$, i.e. when $V_{3(1)}^2 (= V_{1(1)}^2 = V_{2(1)}^2)$ is Cramer's V^2 . Moreover, we note that for an arbitrary $\lambda (\neq 1)$, the value of $V_{3(\lambda)}^2$ lies between $V_{1(\lambda)}^2$ and $V_{2(\lambda)}^2$.

3. Approximate confidence intervals for measures

Assume that the observed frequencies $\{n_{ij}\}$ have a multinomial distribution. We can find an approximate standard error and large-sample confidence interval for $V_{t(\lambda)}^2$ ($t = 1, 2, 3$) using the delta method, which is described in e.g. Bishop et al. (1975, Sec.14.6) and Agresti (1990, p. 57, Chap.12). The sample version of $V_{t(\lambda)}^2$, i.e. $\hat{V}_{t(\lambda)}^2$, is given by $V_{t(\lambda)}^2$ with $\{p_{ij}\}$ replaced by $\{\hat{p}_{ij}\}$, where $\hat{p}_{ij} = n_{ij}/n$. Using the delta method, $\sqrt{n}(\hat{V}_{t(\lambda)}^2 - V_{t(\lambda)}^2)$, $t = 1, 2, 3$, has asymptotically (as $n \rightarrow \infty$) a normal distribution with mean zero and variance $\sigma^2[V_{t(\lambda)}^2]$. See the Appendix for the values of $\sigma^2[V_{t(\lambda)}^2]$.

We note that when $\lambda = 1$, the variance $\sigma^2[V_{t(1)}^2]$ is identical to the variance $\sigma^2[V^2]$ of Cramer's V^2 (see Bishop et al., 1975, Sec.14.6). Let $\hat{\sigma}[V_{t(\lambda)}^2]$ denote $\sigma[V_{t(\lambda)}^2]$ with $\{p_{ij}\}$ replaced by $\{\hat{p}_{ij}\}$. Then $\hat{\sigma}[V_{t(\lambda)}^2]/\sqrt{n}$ ($t = 1, 2, 3$) is an estimated standard error for $\hat{V}_{t(\lambda)}^2$, and $\hat{V}_{t(\lambda)}^2 \pm z_{p/2} \hat{\sigma}[V_{t(\lambda)}^2]/\sqrt{n}$ is an approximate 100(1 - p) percent confidence interval for $V_{t(\lambda)}^2$, where $z_{p/2}$ is the percentage point from the standard normal distribution corresponding to a two-tail probability equal to p .

4. Examples

Example 1.

Consider the data in Table 1, taken from Andersen (1994, p. 358). These are data from a Danish Welfare Study which describes the cross-classification of alcohol consumption and social rank. Alcohol consumption in the contingency table is grouped according to the number of "units" consumed per day. A unit is typically a beer, half a bottle of wine or 2 cl of 40% alcohol. By applying the measure $V_{1(\lambda)}^2$, we consider to what degree prediction of alcohol consumption can be improved when the social rank of an individual is known. Since the confidence intervals for $V_{1(\lambda)}^2$ do not contain zero for any λ (see Table

2a) the indication is that there is a structure of association between alcohol consumption and social rank.

We investigate the strength of association in more detail, using the estimated measure $\hat{V}_{1(\lambda)}^2$ (see Table 2a). For instance, when $\lambda = 0.8$, the strength of association is estimated to be 0.016 times the complete association. So, in predicting the value of alcohol consumption of an individual, we can predict 1.6% better when we know the social rank of the individual than when we do not. Also we note that the power-divergence statistic $W_{(\lambda)}^2$ for testing the independence indicates rejection of the independence hypothesis for these data (see Table 2b).

Example 2.

Consider the data in Table 3, taken from Read and Cressie (1988, p. 20). These are data on 4831 car accidents which are cross-classified according to accident type and accident severity. By applying the measure $V_{1(\lambda)}^2$, we shall consider to what degree prediction of the accident type may be improved if knowledge regarding the accident severity is available. Since the confidence intervals for the measure $V_{1(\lambda)}^2$ do not contain zero for any λ (see Table 4a), these would indicate that there is a structure of association between accident type and accident severity. Also, from the values of estimated measure $\hat{V}_{1(\lambda)}^2$ (see Table 4a), we see that for instance, when $\lambda = 0.8$, the strength of association is estimated to be 0.125 times the complete association. So, in predicting the accident type of a car, we can predict 12.5% better when we know accident severity than when we do not.

When we compare the strength of association between the data in Tables 1 and 3 by using the measure $\hat{V}_{1(\lambda)}^2$, we can see that the degree is greater for Table 3 than for Table 1 because for each $\lambda (\geq 0)$ the values in the confidence interval for $V_{1(\lambda)}^2$ are greater in Table 3 than in Table 1. Moreover, we note that the

power-divergence statistic $W_{(\lambda)}^2$ for testing independence indicates rejection of the independence hypothesis for the data in Table 3 (see Table 4b).

Example 3.

Consider the data in Table 5, taken from Stuart (1955). These are data on unaided distance vision of 7477 women aged 30 to 39 employed in Royal Ordnance factories in Britain from 1943 to 1946. Note that these data have been analyzed by many statisticians - see Tomizawa (1993). Since right eye grade and left eye grade are similar classifications, we shall apply the measure $V_{3(\lambda)}^2$. Since the confidence interval for the measure $V_{3(\lambda)}^2$ does not contain zero for any λ (see Table 6a), the indication is that there is a structure of association between a woman's right eye grade and her left eye grade.

Also, from the values of the estimated measure $\hat{V}_{3(\lambda)}^2$ (see Table 6a), we see that, for instance, when $\lambda = 1.0$ the strength of association between the right eye grade and the left eye grade is estimated to be 0.361 times the complete association. So, when we want to predict a woman's right eye grade and her left eye grade, we can predict it 36.1% better when we use the full distribution $\{\hat{p}_{ij}\}$ than when we use only the marginal distributions $\{\hat{p}_{i.}\}$ and $\{\hat{p}_{.j}\}$. We also note that the power-divergence statistic $W_{(\lambda)}^2$ for testing independence indicates rejection of the independence hypothesis for these data (see Table 6b).

5. Concluding Remarks

The measure $V_{t(\lambda)}^2$ ($t = 1, 2, 3$) always ranges between 0 and 1 independently of the dimensions r and c and the sample size n . Therefore, $V_{t(\lambda)}^2$ may be useful for comparing the strength of association between the row and column variables in several tables. The measure $V_{t(\lambda)}^2$ ($t = 1, 2, 3$) would be useful when we want to see the relative magnitude of the degree of

association between row and column variables to the degree of complete association. Especially, $V_{1(\lambda)}^2$ ($V_{2(\lambda)}^2$) would be useful when the row and column variables are the response (explanatory) and explanatory (response) variables, respectively while $V_{3(\lambda)}^2$ would be useful when explanatory and response variables are not defined.

For the measure $\hat{V}_{t(\lambda)}^2$ ($t = 1, 2, 3$), the analyst may be interested in which value of λ is preferred for a given table. However, it seems difficult to discuss this. It seems to be important (and safe) that in comparing the strength of association in several tables, the analyst calculates the values of $\hat{V}_{t(\lambda)}^2$ for various values of λ and discusses the strength of association in terms of them (rather than calculating $\hat{V}_{t(\lambda)}^2$ for only one specified value of λ , e.g. calculating only Cramer's $\hat{V}^2$). For example, consider the artificial data in Tables 7a and 7b. We see from Table 7c that the value of $\hat{V}_{3(0)}^2$ is greater for Table 7a than for Table 7b, but the value of $\hat{V}_{3(1)}^2$ (Cramer's $\hat{V}^2$) is less for Table 7a than for Table 7b, though the differences are slight in these cases. So, for these cases, it may be impossible to conclude by using $\hat{V}_{3(\lambda)}^2$ for which of Tables 7a and 7b the strength of association is greater. But generally, for comparison between two tables, it would be possible to draw a conclusion if $\hat{V}_{3(\lambda)}^2$ is, for every λ , greater for one table than for the other table. So, for comparison in several tables, we recommend calculating $\hat{V}_{3(\lambda)}^2$ for various λ in addition to Cramer's $\hat{V}^2$. These remarks apply also to the measures $\hat{V}_{1(\lambda)}^2$ and $\hat{V}_{2(\lambda)}^2$.

It may seem to many readers that the power-divergence statistic $W_{(\lambda)}^2$ (including the likelihood ratio chi-squared statistic $G^2 (= W_{(0)}^2)$ and Pearson's chi-squared statistic $X^2 (= W_{(1)}^2)$ as special cases) is also an information measure and, moreover, that $W_{(\lambda)}^2/n$ would be a reasonable measure of strength of association. However, the measure $\hat{V}_{t(\lambda)}^2$ ($t = 1, 2, 3$) would be preferable to $W_{(\lambda)}^2/n$ (for $\lambda \geq 0$) for comparing the strength of association in

several tables. This is because the range of $W_{(\lambda)}^2/n$ (for $\lambda \geq 0$) depends on the marginal proportions, as $0 \leq W_{(\lambda)}^2/n \leq 2\hat{K}_1^{(\lambda)}$, $0 \leq W_{(\lambda)}^2/n \leq 2\hat{K}_2^{(\lambda)}$, or $0 \leq W_{(\lambda)}^2/n \leq (\hat{K}_1^{(\lambda)} + \hat{K}_2^{(\lambda)})$, while $\hat{V}_{t(\lambda)}^2(t = 1, 2, 3)$ is always between 0 and 1, independent of the marginal proportions. For the same reason, the measure $\hat{V}_{t(\lambda)}^2(t = 1, 2, 3)$ would be preferable to $W_{(\lambda)}^2$ (for $\lambda \geq 0$).

We also point out that when one wants to compare the strength of association in several tables, it may be dangerous to use the test statistic $W_{(\lambda)}^2$. This is because it may happen that the value of $\hat{V}_{t(\lambda)}^2(t = 1, 2, 3)$ is greater for table A than for table B, but the value of $W_{(\lambda)}^2$ is less for table A than for table B. For example, consider the artificial data in Tables 8a and 8b. Then we see from Tables 8c and 8d that for each $\lambda \geq 0$ the value of $\hat{V}_{3(\lambda)}^2$ is greater for Table 8a than for Table 8b, but the value of $W_{(\lambda)}^2$ is less for Table 8a than for Table 8b. Thus, the measure $\hat{V}_{t(\lambda)}^2(t = 1, 2, 3)$, rather than the test statistic $W_{(\lambda)}^2$, should be used when we want to compare the relative magnitudes of the strength of association to the degree of complete association in several tables. The test statistic $W_{(\lambda)}^2$ should be used only for testing goodness-of-fit of the independence model.

For analyzing the strength of association between the row and column variables, we first should check whether or not independence holds by using a test statistic, such as $W_{(\lambda)}^2$. Then, if it is judged that there is a structure of association, the next step would be to measure the strength of association by using $\hat{V}_{t(\lambda)}^2(t = 1, 2, 3)$. However, if it is judged that there is independence in the table, then it may not be meaningful to use the measure $\hat{V}_{t(\lambda)}^2$. Furthermore, we note that the measure $\hat{V}_{t(\lambda)}^2(t = 1, 2, 3)$ is invariant under arbitrary permutations of categories. Therefore it is possible to apply these measures for analyzing data that are on a nominal or ordinal scale.

We observe that (i) the estimate of the strength of association should be considered in terms of an approximate confidence interval for the measure $V_{t(\lambda)}^2$ ($t = 1, 2, 3$) and not in terms of $\hat{V}_{t(\lambda)}^2$ itself; (ii) the measure would be useful for describing relative magnitudes (of strength of association) rather than absolute magnitudes, and (iii) $\hat{V}_{t(\lambda)}^2$ should not be used for testing goodness-of-fit of the independence model.

6. Discussion

The measure $V_{3(\lambda)}^2$ may be expressed as

$$V_{3(\lambda)}^2 = \frac{2V_{1(\lambda)}^2 V_{2(\lambda)}^2}{V_{1(\lambda)}^2 + V_{2(\lambda)}^2} \quad \text{for } \lambda \geq 0.$$

$V_{3(\lambda)}^2$ is the harmonic mean of $V_{1(\lambda)}^2$ and $V_{2(\lambda)}^2$. A referee suggested the measure (instead of $V_{3(\lambda)}^2$)

$$VG_{(\lambda)}^2 = \sqrt{V_{1(\lambda)}^2 V_{2(\lambda)}^2} = \frac{I^{(\lambda)}(\{p_{ij}\}; \{p_{i.} p_{.j}\})}{\sqrt{K_1^{(\lambda)} K_2^{(\lambda)}}} \quad \text{for } \lambda \geq 0.$$

This is the geometric mean of $V_{1(\lambda)}^2$ and $V_{2(\lambda)}^2$. Note that when $\lambda = 1$ for the $r \times c$ table, $VG_{(1)}^2$ is identical to Tshuprow's coefficient T^2 (see Section 1). The measures $V_{3(\lambda)}^2$ and $VG_{(\lambda)}^2$ both have the properties that they are on the interval $[0, 1]$, equal to 1 only in the case of complete association and equal to 0 in the case of independence. Also, it is easily seen that for any fixed λ , $\min(V_{1(\lambda)}^2, V_{2(\lambda)}^2) \leq V_{3(\lambda)}^2 \leq VG_{(\lambda)}^2 \leq \max(V_{1(\lambda)}^2, V_{2(\lambda)}^2)$, and that the equality holds if and only if $V_{1(\lambda)}^2 = V_{2(\lambda)}^2$.

Table 9 gives the estimated values of $V_{3(\lambda)}^2$ and $VG_{(\lambda)}^2$ applied to Examples 1, 2, and 3. For these data, the differences between the estimated values of $V_{3(\lambda)}^2$ and $VG_{(\lambda)}^2$ are very slight. It seems difficult to discuss theoretically which of measures $V_{3(\lambda)}^2$ and $VG_{(\lambda)}^2$ is suitable for representing the strength of association.

Consider the artificial data in Tables 10a, 10b and 10c. First, we shall consider the comparison of measures applied to the data in Tables 10a and 10b. We see from Table 10d that for each λ the value of $\hat{V}_{1(\lambda)}^2$ for Table 10a is equal to that for Table 10b, but the value of $\hat{V}_{2(\lambda)}^2$ is greater for Table 10a than for Table 10b. Recall that the measure $V_{1(\lambda)}^2$ ($V_{2(\lambda)}^2$) indicates to what degree prediction of the row (column) category is improved by knowledge of the column (row) category. Thus, for predicting row values from column values $V_{1(\lambda)}^2$ may be useful while $V_{2(\lambda)}^2$ may be useful for predicting column values from row values. In addition, we note that $V_{1(\lambda)}^2$ would not be useful in measuring the extent to which prediction of the value of a variable is improved by knowledge of another variable.

Finally, we consider a comparison of the measures applied to the data in Tables 10a and 10c. We see from Table 10d that for each λ , the value of $\hat{V}_{2(\lambda)}^2$ for Table 10a is equal to that for Table 10c, but the value of $\hat{V}_{1(\lambda)}^2$ is greater for Table 10a than for Table 10c. Similarly to the comparison of Tables 10a and 10b, we can see that $V_{2(\lambda)}^2$ would not be useful in measuring the extent to which prediction of the value of a variable is improved by knowledge of another variable. So, for a table in which, e.g. the explanatory and response variables are not defined, the strength of association should be measured by $V_{3(\lambda)}^2$ or $VG_{(\lambda)}^2$ (instead of $V_{1(\lambda)}^2$ and $V_{2(\lambda)}^2$).

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Appendix

Using the delta method, $\sqrt{n}(\hat{V}_{t(\lambda)}^2 - V_{t(\lambda)}^2)$, $t = 1, 2, 3$, has asymptotic variance

$$(i) \sigma^2[V_{1(\lambda)}^2] = \frac{1}{(L_1^{(\lambda)})^2} \left[\sum_{i=1}^r \sum_{j=1}^c \left(\Delta_{1ij}^{(\lambda)} \right)^2 p_{ij} - \left(\sum_{s=1}^r \sum_{t=1}^c \Delta_{1st}^{(\lambda)} p_{st} \right)^2 \right],$$

where

$$\Delta_{1ij}^{(\lambda)} = \begin{cases} H_{ij}^{(\lambda)} - (1-\lambda)V_{1(\lambda)}^2 p_i^{-\lambda} & \text{for } \lambda > 0, \\ \log\left(\frac{p_{ij}}{p_i \cdot p_j}\right) - 1 + V_{1(0)}^2 (\log p_i + 1) & \text{for } \lambda = 0, \end{cases}$$

$$H_{ij}^{(\lambda)} = (\lambda+1) \left(\frac{p_{ij}}{p_i \cdot p_j}\right)^\lambda - \lambda \sum_{t=1}^c \frac{p_{it}}{p_i} \left(\frac{p_{it}}{p_i \cdot p_t}\right)^\lambda - \lambda \sum_{s=1}^r \frac{p_{sj}}{p_j} \left(\frac{p_{sj}}{p_s \cdot p_j}\right)^\lambda,$$

$$L_1^{(\lambda)} = \begin{cases} \sum_{i=1}^r p_i^{1-\lambda} - 1 & \text{for } \lambda > 0, \\ -\sum_{i=1}^r p_i \log p_i & \text{for } \lambda = 0, \end{cases}$$

$$(ii) \sigma^2[V_{2(\lambda)}^2] = \frac{1}{(L_2^{(\lambda)})^2} \left[\sum_{i=1}^r \sum_{j=1}^c \left(\Delta_{2ij}^{(\lambda)}\right)^2 p_{ij} - \left(\sum_{s=1}^r \sum_{t=1}^c \Delta_{2st}^{(\lambda)} p_{st} \right)^2 \right],$$

where

$$\Delta_{2ij}^{(\lambda)} = \begin{cases} H_{ij}^{(\lambda)} - (1-\lambda)V_{2(\lambda)}^2 p_j^{-\lambda} & \text{for } \lambda > 0, \\ \log\left(\frac{p_{ij}}{p_i \cdot p_j}\right) - 1 + V_{2(0)}^2 (\log p_j + 1) & \text{for } \lambda = 0, \end{cases}$$

$$L_2^{(\lambda)} = \begin{cases} \sum_{j=1}^c p_j^{1-\lambda} - 1 & \text{for } \lambda > 0, \\ -\sum_{j=1}^c p_j \log p_j & \text{for } \lambda = 0, \end{cases}$$

$$(iii) \sigma^2[V_{3(\lambda)}^2] = \frac{1}{(L_1^{(\lambda)} + L_2^{(\lambda)})^2} \left[\sum_{i=1}^r \sum_{j=1}^c \left(\Delta_{3ij}^{(\lambda)}\right)^2 p_{ij} - \left(\sum_{s=1}^r \sum_{t=1}^c \Delta_{3st}^{(\lambda)} p_{st} \right)^2 \right],$$

where

$$\Delta_{3ij}^{(\lambda)} = \begin{cases} 2H_{ij}^{(\lambda)} - (1-\lambda)V_{3(\lambda)}^2 (p_i^{-\lambda} + p_j^{-\lambda}) & \text{for } \lambda > 0 \\ 2 \log\left(\frac{p_{ij}}{p_i \cdot p_j}\right) - 2 + V_{3(0)}^2 (\log p_i + \log p_j + 2) & \text{for } \lambda = 0. \end{cases}$$

Table 1. Association between alcohol consumption and social rank

Units of alcohol consumed per day	Social rank group				Total
	I-II	III	IV	V	
Under 1	98	338	484	484	1404
1-2	235	406	588	385	1614
More than 2	123	144	191	137	595
Total	456	888	1263	1006	3613

Source: Andersen (1994, p. 358)

Table 2. (a) Estimate of measure $V_{1(\lambda)}^2$, estimated approximate standard error for $\hat{V}_{1(\lambda)}^2$, and approximate 95% confidence interval for $V_{1(\lambda)}^2$, and (b) value of power-divergence statistic $W_{(\lambda)}^2$, applied to Table 1

(a) For $\hat{V}_{1(\lambda)}^2$

Values of λ	$\hat{V}_{1(\lambda)}^2$	Standard error	Confidence interval
0.0	0.015	0.003	(0.010, 0.020)
0.4	0.016	0.003	(0.010, 0.022)
0.8	0.016	0.003	(0.010, 0.022)
1.0	0.015	0.003	(0.010, 0.021)
1.4	0.013	0.003	(0.008, 0.018)
1.8	0.011	0.002	(0.007, 0.015)
2.0	0.009	0.002	(0.006, 0.013)

(b) For $W_{(\lambda)}^2$

Values of λ	$W_{(\lambda)}^2$
0.0	110.9
0.4	110.1
0.8	109.9
1.0	110.0
1.4	110.6
1.8	111.7
2.0	112.5

Table 3. The 4831 car accidents

Accident type	Accident severity			Total
	Not severe	Moderately severe	Severe	
Rollover	2365	944	412	3721
Not rollover	249	585	276	1110
Total	2614	1529	688	4831

Source: Cressie and Read (1988, p. 20)

Table 4. (a) Estimate of measure $V_{1(\lambda)}^2$, estimated approximate standard error for $\hat{V}_{1(\lambda)}^2$, and approximate 95% confidence interval for $V_{1(\lambda)}^2$, and (b) value of power-divergence statistic $W_{(\lambda)}^2$, applied to Table 3

(a) For $\hat{V}_{1(\lambda)}^2$

Values of λ	$\hat{V}_{1(\lambda)}^2$	Standard error	Confidence interval
0.0	0.116	0.009	(0.099, 0.133)
0.4	0.127	0.009	(0.109, 0.146)
0.8	0.125	0.009	(0.107, 0.144)
1.0	0.121	0.009	(0.103, 0.138)
1.4	0.107	0.008	(0.091, 0.123)
1.8	0.089	0.007	(0.075, 0.103)
2.0	0.080	0.007	(0.067, 0.093)

(b) For $W_{(\lambda)}^2$

Values of λ	$W_{(\lambda)}^2$
0.0	602.1
0.4	590.2
0.8	584.2
1.0	583.3
1.4	585.3
1.8	592.5
2.0	598.0

Table 5. Unaided distance vision data of British women

Right eye grade	Left eye grade				Total
	Best (1)	Second (2)	Third (3)	Worst (4)	
Best (1)	1520	266	124	66	1976
Second (2)	234	1512	432	78	2256
Third (3)	117	362	1772	205	2456
Worst (4)	36	82	179	492	789
Total	1907	2222	2507	841	7477

Source: Stuart (1955)

Table 6. (a) Estimate of measure $V_{3(\lambda)}^2$, estimated approximate standard error for $\hat{V}_{3(\lambda)}^2$, and approximate 95% confidence interval for $V_{3(\lambda)}^2$, and (b) value of power-divergence statistic $W_{(\lambda)}^2$, applied to Table 5

(a) For $\hat{V}_{3(\lambda)}^2$

Values of λ	$\hat{V}_{3(\lambda)}^2$	Standard error	Confidence interval
0.0	0.338	0.008	(0.324, 0.353)
0.4	0.369	0.008	(0.353, 0.385)
0.8	0.369	0.009	(0.352, 0.386)
1.0	0.361	0.009	(0.344, 0.378)
1.4	0.334	0.009	(0.316, 0.351)
1.8	0.296	0.009	(0.278, 0.315)
2.0	0.276	0.009	(0.257, 0.294)

(b) For $W_{(\lambda)}^2$

Values of λ	$W_{(\lambda)}^2$
0.0	6671.5
0.4	6996.9
0.8	7635.9
1.0	8096.9
1.4	9382.9
1.8	11326.4
2.0	12639.2

Table 7. Artificial data (Tables 7a and 7b) and corresponding values of $\hat{V}_{3(\lambda)}^2$ (n is sample size)

(a) $n = 1928$			(b) $n = 1751$		
1264	82	78	304	201	210
80	88	76	56	329	198
84	86	90	38	45	370

(c) Value of $\hat{V}_{3(\lambda)}^2$		
Values of λ	for Table 7a	for Table 7b
0.0*	0.197	0.150
0.4*	0.199	0.166
0.8*	0.176	0.168
1.0	0.159	0.165
1.4	0.123	0.153
1.8	0.089	0.137
2.0	0.075	0.128
2.4	0.051	0.110

* indicates that the value of $\hat{V}_{3(\lambda)}^2$ is greater for Table 7a than for Table 7b

Table 8. Artificial data (Tables 8a and 8b), the corresponding values of $\hat{V}_{3(\lambda)}^2$ and the values of power-divergence statistic $W_{(\lambda)}^2$ (n is sample size)

(a) $n = 294$			(b) $n = 820$		
241	6	7	600	31	29
7	6	7	27	29	27
8	7	5	29	25	23

(c) For $\hat{V}_{3(\lambda)}^2$		
Values of λ	for Table 8a	for Table 8b
0.0	0.251	0.204
0.4	0.244	0.203
0.8	0.202	0.173
1.0	0.176	0.154
1.4	0.125	0.113
1.8	0.084	0.079
2.0	0.069	0.064
2.4	0.045	0.042

(d) For $W_{(\lambda)}^2$		
Values of λ	for Table 8a	for Table 8b
0.0	71.4	211.8
0.4	79.7	222.3
0.8	93.6	240.1
1.0	103.3	252.3
1.4	130.8	284.7
1.8	173.9	330.9
2.0	203.9	360.6
2.4	289.3	437.2

Table 9. Comparisons between the estimated values of measures $V_{t(\lambda)}^2$, $t = 1, 2, 3$ and $VG_{(\lambda)}^2$ applied to Examples 1, 2 and 3.

Value of λ	Examples	$\hat{V}_{1(\lambda)}^2$	$\hat{V}_{2(\lambda)}^2$	$\hat{V}_{3(\lambda)}^2$	$\hat{VG}_{(\lambda)}^2$
0	1	0.015	0.012	0.013	0.013
	2	0.116	0.064	0.082	0.086
	3	0.339	0.338	0.338	0.338
0.4	1	0.016	0.012	0.014	0.014
	2	0.127	0.068	0.089	0.090
	3	0.369	0.368	0.369	0.369
0.8	1	0.016	0.011	0.013	0.013
	2	0.125	0.064	0.085	0.090
	3	0.369	0.369	0.369	0.369
1.0	1	0.015	0.010	0.012	0.012
	2	0.121	0.060	0.080	0.085
	3	0.361	0.361	0.361	0.361
1.4	1	0.013	0.008	0.010	0.010
	2	0.107	0.050	0.068	0.073
	3	0.333	0.335	0.334	0.334

Table 10. Artificial data (Tables 10a, 10b and 10c) and the corresponding estimated values of $V_{t(\lambda)}^2$, $t = 1, 2, 3$, and $VG_{(\lambda)}^2$.

(a)		(b)			(c)	
8	4	8	2	2	4	2
0	12	0	6	6	4	2
					0	12

(d) Estimated measures					
Value of λ	Tables	$\hat{V}_{1(\lambda)}^2$	$\hat{V}_{2(\lambda)}^2$	$\hat{V}_{3(\lambda)}^2$	$\hat{VG}_{(\lambda)}^2$
0	(a)	0.459	0.500	0.479	0.479
	(b)	0.459	0.290	0.355	0.365
	(c)	0.306	0.500	0.380	0.391
0.6	(a)	0.490	0.511	0.501	0.501
	(b)	0.490	0.271	0.349	0.365
	(c)	0.279	0.511	0.361	0.378
1.0	(a)	0.500	0.500	0.500	0.500
	(b)	0.500	0.250	0.333	0.354
	(c)	0.250	0.500	0.333	0.354